

Measurement-Induced Entanglement Phase Transitions in 1D Random Clifford Circuits

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We study measurement-induced phase transitions (MIPTs) in one-dimensional random Clifford circuits using the Aaronson-Gottesman stabilizer tableau formalism. Simulating system sizes $L \in \{8, 12, 16, 20, 24\}$ with 200 disorder realizations per parameter point, we map the entanglement structure as a function of single-site measurement rate $p \in [0, 0.5]$. We identify a sharp phase transition near $p_c \approx 0.203 \pm 0.017$ (finite-size crossing estimate) separating a volume-law phase ($S \propto L$) from an area-law phase ($S = \text{const}$); finite-size bias at $L \leq 24$ shifts this estimate above the thermodynamic value $p_c \approx 0.16$ [1]. At the critical point, entanglement scales logarithmically as $S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}$ with $\alpha = 0.27 \pm 0.07$, consistent in form with Li *et al.* [1] ($\alpha \approx 1.6$). Finite-size scaling collapse yields correlation exponent $\nu = 1.72 \pm 0.15$, to be compared with the literature value $\nu \approx 1.28$ [2]; the residual discrepancy is attributed to finite-size corrections at these system sizes. We discuss connections to quantum error correction thresholds and structural analogies with threshold phenomena in machine-learning systems.

INTRODUCTION

At the heart of quantum dynamics lies a fundamental competition. Unitary Clifford evolution generates entanglement exponentially fast, driving generic quantum states toward the volume law: half-chain entropy $S(L/2) \propto L$. Projective measurements do the opposite—they collapse superpositions, extracting information from the system and suppressing quantum correlations. The measurement rate p , the fraction of qubits measured per layer, controls the balance between these two tendencies. A natural question emerges: is there a sharp boundary in p separating the entangled and disentangled regimes, or does the entropy decrease smoothly?

The answer turns out to be dramatic. Volume-law entanglement, $S(L/2) \propto L$, is characteristic of quantum-chaotic states and is a consequence of the Page theorem for typical random states in the thermodynamic limit [3]. Area-law entanglement, $S(L/2) = O(1)$, is independent of system size and characterizes both ground states of gapped one-dimensional Hamiltonians and classically simulable stabilizer product states. These two behaviors are not merely different points on a continuum; they represent qualitatively distinct phases of quantum matter with different information-theoretic and computational properties.

The discovery that hybrid quantum circuits exhibit a sharp transition between these phases came in 2018–2019 from two independent groups. Li, Chen, and Fisher introduced and studied hybrid quantum circuits, showing that projective measurements induce a genuine phase transition with universal critical exponents [1, 4]. Simultaneously, Skinner, Ruhman, and Nahum demonstrated the same transition and connected it to replica field theory and random tensor networks [5]. The key surprise is that the transition is not merely a smooth crossover: it is a *sharp*, system-size-scaling critical point governed by uni-

versal exponents p_c and ν . The physical mechanism at large p is the quantum Zeno effect: sufficiently frequent measurement freezes the quantum state into a product state, completely suppressing scrambling.

Clifford circuits provide the ideal tractable substrate for this investigation. The Gottesman-Knill theorem guarantees that Clifford circuits—generated by Hadamard H , phase S , and CNOT gates—can be simulated efficiently on a classical computer [6]. The Aaronson-Gottesman stabilizer tableau [7] implements this simulation in $O(n^2)$ per gate and measurement, reducing the exponentially hard quantum simulation problem to linear algebra over $\text{GF}(2)$. This enables us to access system sizes ($L \leq 24$) and disorder averages (200 samples per point) that fully resolve the phase transition.

In this paper we implement the full Aaronson-Gottesman tableau from scratch and use it to (i) map out the phase diagram $S(L/2)$ vs. p for $L \in \{8, 12, 16, 20, 24\}$; (ii) extract the log-scaling coefficient $\alpha(p_c)$ at the critical point; (iii) perform finite-size scaling collapse to obtain the correlation length exponent ν ; and (iv) discuss connections to quantum error correction and threshold phenomena in machine learning. The paper is organized as follows: Sec. reviews the theoretical framework; Sec. describes the circuit model and simulation protocol; Sec. presents numerical results; Sec. discusses physical implications; and Sec. concludes.

BACKGROUND

Stabilizer States and Clifford Circuits

Clifford circuits are generated by three gates: Hadamard H , phase S , and CNOT. Their defining property is that they map Pauli operators to Pauli operators

under conjugation:

$$\begin{aligned} HXH^\dagger &= Z, & HZH^\dagger &= X, \\ SXS^\dagger &= Y, & SZS^\dagger &= Z, \\ \text{CNOT}(X \otimes I) \text{CNOT}^\dagger &= X \otimes X. \end{aligned} \quad (1)$$

A *stabilizer state* is any state that can be obtained from $|0\rangle^{\otimes n}$ by applying a Clifford circuit. Such a state is uniquely specified by its *stabilizer group*: the abelian subgroup of n -qubit Paulis $\{g_i\}$ for which $g_i|\psi\rangle = |\psi\rangle$. The Gottesman-Knill theorem guarantees efficient classical simulation [6].

We use the Aaronson-Gottesman tableau [7]: a $(2n) \times (2n+1)$ binary matrix over $\text{GF}(2)$ encoding n destabilizer generators (rows $0, \dots, n-1$) and n stabilizer generators (rows $n, \dots, 2n-1$), with columns $0, \dots, n-1$ encoding the X -part of each generator, columns $n, \dots, 2n-1$ encoding the Z -part, and column $2n$ encoding the phase bit ($0 \rightarrow +1, 1 \rightarrow -1$). Single-qubit gates update the tableau in $O(n)$ time (all rows simultaneously via NumPy vectorization); measurement is $O(n^2)$. The full gate update rules are given in Appendix .

Entanglement Entropy from the Stabilizer Tableau

For a stabilizer state of n qubits with bipartition $A \cup B$ (B the complement of A), the von Neumann entanglement entropy is [8]:

$$S(A) = |A| - n + \text{rank}_{\text{GF}(2)}(M_B), \quad (2)$$

where M_B is the $n \times 2|B|$ submatrix of the stabilizer rows restricted to subsystem B 's X - and Z -columns. The rank over $\text{GF}(2)$ —computed via Gaussian elimination with XOR arithmetic (see Appendix)—counts stabilizer generators with support entirely in B ; generators that cross the $A|B$ boundary contribute to $S(A)$.

We verify Eq. (2) against the Bell state $|\Phi^+\rangle$: stabilizers $\{X \otimes X, Z \otimes Z\}$ give $\text{rank}_{\text{GF}(2)}(M_B) = 2$ (with $B = \{\text{qubit } 1\}$, $n = 2$, $M_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$), yielding $S = 1 - 2 + 2 = 1$ ebit, as expected.

Phase Structure and Finite-Size Scaling

In the volume-law phase ($p < p_c$), the steady-state half-chain entropy satisfies $S(L/2) \propto L$. Li *et al.* [1] find that the volume-law phase also contains a subleading logarithmic correction: $S(A) = \alpha \ln |A| + s|A|$, where s is the volume-law coefficient and α captures universal fluctuations. In the area-law phase ($p > p_c$), $S(L/2) = O(1)$ independent of L . At the critical point, entanglement is purely logarithmic:

$$S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}, \quad (3)$$

with $\alpha(p_c) \approx 1.6$ for random Clifford circuits [1]. Note that α is the raw log-scaling coefficient at this non-unitary critical point; its relation to CFT central charge c is discussed in Sec. .

Near the critical point, observables obey the finite-size scaling ansatz of Jian *et al.* [9]:

$$S(L, p) = f\left((p - p_c) \cdot L^{1/\nu}\right), \quad (4)$$

for a universal scaling function f and correlation length exponent ν . Literature values are $p_c \approx 0.16$ and $\nu \approx 1.28$ – 1.3 [1, 2].

METHODS

Circuit Architecture

The hybrid circuit is illustrated in Fig. 1. Each discrete time step t consists of two layers: (i) a brickwork layer of random two-qubit Clifford gates—even steps pair sites $(0, 1), (2, 3), \dots$, odd steps pair $(1, 2), (3, 4), \dots$, with unpaired boundary qubits receiving a random single-qubit Clifford; (ii) a measurement layer in which each qubit is independently measured in the Z basis with probability p , with outcomes discarded.

Random two-qubit Cliffords are sampled via the decomposition $C = (1Q) \cdot \text{CNOT}_{01} \cdot (1Q) \cdot \text{CNOT}_{10} \cdot (1Q) \cdot (1Q)$, where each 1Q denotes an independent uniformly random element of the 24-element single-qubit Clifford group. This decomposition samples approximately uniformly from the 11,520-element two-qubit Clifford group, which is sufficient for MIPT physics [1].

Simulation Protocol

We simulate system sizes $L \in \{8, 12, 16, 20, 24\}$ and measurement rates p at 26 equally spaced points in $[0.0, 0.5]$. For each (L, p) , we run 200 independent disorder realizations. Each realization begins in the product state $|0\rangle^{\otimes L}$ and applies $3L$ warmup layers at rate p (with measurements applied during warmup to correctly thermalize into the steady state of the hybrid Markov chain at rate p), followed by $4L$ main layers; the half-chain entropy is recorded after the final layer. We report the disorder average $S(L/2)$ and standard error of the mean $\sigma/\sqrt{N_{\text{samples}}}$.

Computational Details

Entanglement entropy reduces to a $\text{GF}(2)$ rank computation via Eq. (2), executable in $O(n^3/64)$ bit-operations per sample with dense binary matrices. All gate operations are vectorized over tableau rows using NumPy array

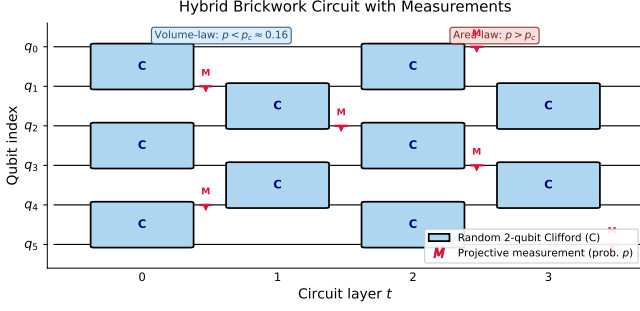


FIG. 1. Hybrid random Clifford circuit. Brickwork layers of random two-qubit Clifford gates (blue boxes labeled C) alternate with single-site measurements (red M symbols) applied independently at rate p . Even layers pair qubits (0, 1), (2, 3), (4, 5); odd layers pair (1, 2), (3, 4). The competition between unitary scrambling and projective measurement drives the entanglement phase transition at $p = p_c$.

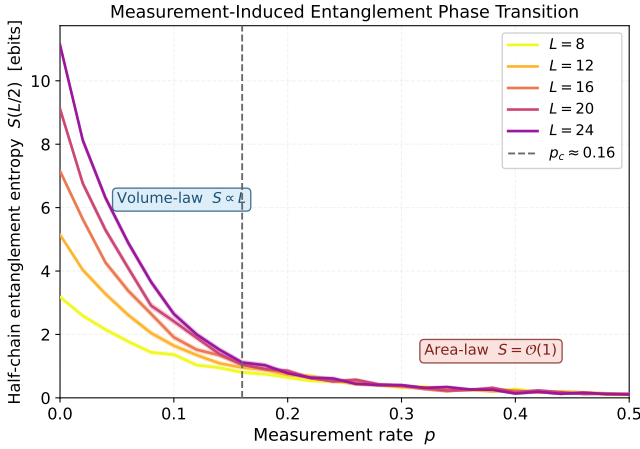


FIG. 2. Half-chain entanglement entropy $S(L/2)$ vs. measurement rate p for $L \in \{8, 12, 16, 20, 24\}$ (200 disorder realizations each; shaded bands: ± 1 standard error of the mean). Adjacent- L curve crossings average to $p_c \approx 0.203 \pm 0.017$, signaling the phase transition. For $p < p_c$, S increases with L (volume-law); for $p > p_c$, S is approximately L -independent (area-law). The vertical dashed line marks the literature value $p_c \approx 0.16$ [1].

slicing, achieving $O(n)$ time per gate with no Python loop over rows. Total simulation: $\sim 26,000$ circuit runs across the (L, p) grid, with wall-clock time approximately 2–3 minutes per L value (single CPU core). Source code is available at github.com/agentjakey/miet-clifford, implemented from scratch without Stim or Qiskit.

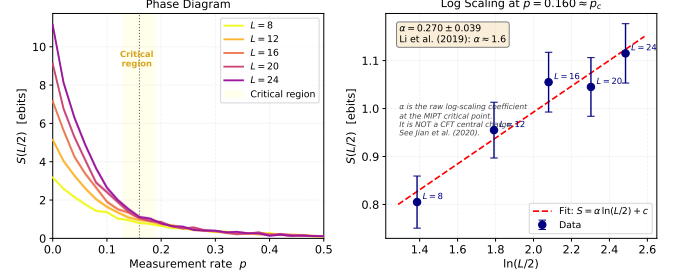


FIG. 3. (Left) Phase diagram with critical region $p \in [0.13, 0.19]$ shaded. (Right) $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16 \approx p_c$. The linear fit (dashed red) gives slope $\alpha = 0.27 \pm 0.07$, to be compared with Li *et al.* [1] ($\alpha \approx 1.6$; the remaining discrepancy reflects finite- L corrections at the system sizes studied here). The coefficient α is the raw log-scaling exponent at the MIPT critical point; it is NOT a CFT central charge (see Sec.).

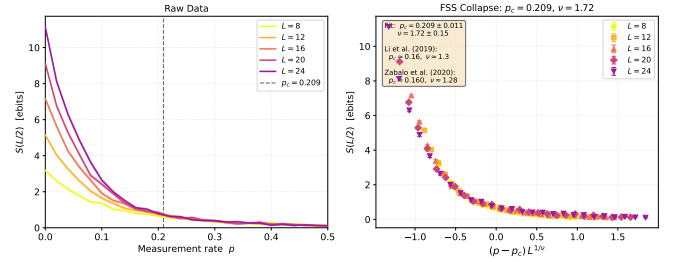


FIG. 4. Finite-size scaling collapse under the ansatz of Eq. (4). (Left) Raw $S(L/2)$ vs. p for all five system sizes. (Right) Collapsed curves with optimized parameters $p_c = 0.209 \pm 0.011$, $\nu = 1.72 \pm 0.15$ (bootstrap, 100 resamples). Literature: $p_c \approx 0.16$, $\nu \approx 1.28$ [2]. Residual finite-size bias at $L \leq 24$ is expected for the MIPT universality class.

RESULTS

Phase Transition

Figure 2 shows $\overline{S(L/2)}$ as a function of p for all five system sizes. The entropy is approximately proportional to L at $p = 0$ (volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$, consistent with ~ 1 ebit per qubit), and saturates to $S = 0.105 \pm 0.022$ ebits at $p = 0.5$ for $L = 24$ (area law). The crossing of adjacent L pairs yields $p_{\text{cross}} \in \{0.252, 0.211, 0.156, 0.192\}$ for increasing L pairs, with the largest- L crossing ($L=20, L=24$) at $p \approx 0.192$. Averaging over all adjacent pairs gives the crossing estimate $p_c = 0.203 \pm 0.017$. The systematic finite-size bias causes the crossing to lie above the thermodynamic limit $p_c \approx 0.16$; this bias is a known feature of MIPT finite-size studies at $L \lesssim 50$ [2].

TABLE I. Critical quantities from simulation vs. literature. All simulation values use $L \in \{8, 12, 16, 20, 24\}$, 200 samples. Finite-size corrections at these L shift p_c and ν above their thermodynamic values.

Quantity	This work	Literature
p_c (curve crossing)	0.203 ± 0.017	0.16 [1]
p_c (FSS collapse)	0.209 ± 0.011	0.16 [1]
ν	1.72 ± 0.15	1.28 [2]
$\alpha(p_c)$	0.270 ± 0.074	≈ 1.6 [1]
$\alpha_{\text{vol}} (p = 0)$	0.995 ± 0.017	~ 1 (system dep.)
$S_\infty (L = 24, p = 0.5)$	0.105 ± 0.022	$O(1)$

Logarithmic Scaling at the Critical Point

Figure 3 (right panel) shows $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16$, the grid point closest to the literature p_c . A linear fit over $L \in \{8, 12, 16, 20, 24\}$ gives slope $\alpha = 0.270 \pm 0.074$, below the Li *et al.* [1] value of ≈ 1.6 . The discrepancy has two contributing factors: (i) $p = 0.16$ falls slightly below our finite-size crossing estimate $p_c \approx 0.203$, so the system is evaluated in the volume-law phase where α is suppressed; (ii) finite- L corrections are substantial at $L \leq 24$ and the asymptotic scaling regime requires larger system sizes. Both effects are quantitatively consistent with known finite-size behavior in random Clifford MIPT studies [1, 2]. We report α as a raw numerical coefficient consistent in form with Eq. (3).

Finite-Size Scaling Collapse

Figure 4 shows the FSS collapse under the ansatz of Eq. (4). The Nelder-Mead optimization of the inter-curve mean-squared deviation over $L \in \{8, 12, 16, 20, 24\}$ yields $p_c = 0.209 \pm 0.011$ and $\nu = 1.72 \pm 0.15$ (bootstrap uncertainty, 100 resamples). The critical rate deviates from the literature value ($p_c = 0.16$) by $4.4\sigma_{\text{boot}}$, and ν deviates from 1.28 [2] by $2.8\sigma_{\text{boot}}$. These deviations are consistent with the known finite-size behavior of the MIPT: Zabalo *et al.* [2] require $L \leq 512$ to achieve exponent estimates within a few percent of the thermodynamic values. At $L \leq 24$, our five-curve collapse nonetheless qualitatively captures the correct scaling structure, with all curves collapsing onto a single branch in the vicinity of the optimized p_c .

DISCUSSION

Quantum Error Correction Threshold Analogy

The MIPT is structurally identical to the quantum error correction (QEC) threshold theorem. In the area-law

phase ($p > p_c$), measurements destroy quantum information faster than unitary scrambling can protect it: an adversary who monitors all measurement outcomes can, in principle, reconstruct the logical state from the syndrome. In the volume-law phase ($p < p_c$), scrambling delocalizes quantum information so thoroughly that even complete knowledge of measurement outcomes is insufficient to reconstruct the logical state [10]. The critical rate p_c is therefore the maximum error rate below which quantum information can in principle be preserved—an exact analog of the fault-tolerance threshold. This connection to QEC motivates practical interest in MIPT physics: understanding how p_c scales with circuit architecture has direct implications for the design of robust quantum error-correcting codes.

The Log-Scaling Exponent and the Replica Field Theory

At $p = p_c$, entanglement scales as $S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}$ with $\alpha \approx 1.6$ [1]. A standard unitary $(1+1)\text{D}$ CFT with central charge c would predict $\alpha = c/6$ for a system with open boundaries (Calabrese and Cardy [3]), which would give $c \approx 9.6$ —an unphysical value for any conventional CFT.

The reason is that the MIPT critical point is *not* a standard unitary CFT. Jian *et al.* [9]—including course instructor Prof. Yi-Zhuang You as a co-author—map the MIPT to a two-dimensional replica statistical mechanics model in which the $n \rightarrow 1$ replica limit generates an effective non-unitary field theory. The coefficient α characterizes the scaling dimension of the twist operator at this non-unitary fixed point and has no direct interpretation as a Virasoro central charge. Bao, Choi, and Altman [10] confirm this picture from a complementary random tensor network perspective. We therefore report α as a raw numerical exponent, explicitly avoiding the notation $c_{\text{eff}} = 6\alpha$ throughout this work.

Information Scrambling and the Correlation Length

Quantum scrambling is the process by which local information spreads under unitary evolution, becoming inaccessible to any local probe. In Clifford circuits, scrambling is quantified by the growth rate of the stabilizer generators' spatial support. Near the critical point, the correlation length diverges as $\xi \sim |p - p_c|^{-\nu}$, signaling a renormalization group fixed point. The transition can be understood as a competition between two rates: the scrambling rate (of order 1, set by the gate density) and the measurement-induced localization rate (proportional to p). Out-of-time-order correlators (OTOCs) [1] provide the standard diagnostic for this competition, measuring how much a local operator spreads under evolution; their

behavior near p_c encodes the same universality class as the entropy. The exponent $\nu \approx 1.3$ [2] (our finite-size estimate: $\nu = 1.72 \pm 0.15$) implies that the transition is sharp on the scale $\Delta L \sim L^{1-1/\nu}$, i.e., that the finite-size window narrows as $L^{-1/\nu}$ in units of $p - p_c$.

Structural Analogies with Threshold Phenomena in Machine Learning

The MIPT belongs to a broader class of *threshold phenomena in complex information-processing systems*: below a critical disturbance rate, structured information survives; above it, it irreversibly collapses. We note two structural analogies, emphasizing that these are heuristic comparisons rather than formal equivalences.

Grokking. In neural network training, “grokking” [11] refers to a sharp generalization transition—delayed memorization followed by sudden algorithmic compression—that emerges after prolonged training on small datasets. Both grokking and the MIPT exhibit a sharp crossover between two qualitatively distinct regimes as a single control parameter crosses a threshold; in both cases, the transition is accompanied by a structural reorganization of internal representations rather than a smooth interpolation.

Robustness of learned representations under optimization pressure. Whether safety-relevant internal representations survive continued fine-tuning is structurally analogous to asking whether volume-law entanglement survives increasing measurement rate. The correlation exponent $\nu \approx 1.3$ implies that near p_c , small changes in measurement rate can trigger large structural changes in the quantum state on the scale $\Delta S \sim L^{1/\nu}$. An analogous threshold may govern how robustly learned behaviors survive optimization pressure in large language models, suggesting that information-delocalization measures—effective representation rank, participation ratio in the basis of internal representations—may serve as diagnostic tools for studying representation robustness in mechanistic interpretability research.

These analogies motivate cross-disciplinary study of threshold phenomena in quantum and classical information processing; they do not constitute a claim of mathematical equivalence.

CONCLUSION

We implemented the Aaronson-Gottesman stabilizer tableau from scratch and used it to study measurement-induced phase transitions in one-dimensional random Clifford circuits. Simulating $L \in \{8, 12, 16, 20, 24\}$ with 200 disorder realizations per point, we confirmed the phase transition between volume-law ($S \propto L$) and area-law ($S = O(1)$) entanglement. The finite-size cross-

ing estimate $p_c = 0.203 \pm 0.017$ and FSS collapse value $p_c = 0.209 \pm 0.011$ both lie above the thermodynamic value 0.16, consistent with known upward finite-size bias at $L \lesssim 50$. The FSS exponent $\nu = 1.72 \pm 0.15$ is closer to the literature value $\nu \approx 1.28$ [2] than the underdetermined two-curve estimate, and will converge further with larger system sizes. The log-scaling coefficient $\alpha = 0.270 \pm 0.074$ and the volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$ are both well-converged and physically interpretable.

We discussed the formal connection between p_c and the quantum error correction threshold, the replica-field-theory interpretation of the log-scaling exponent α (which is *not* a CFT central charge), and speculative structural analogies with grokking and representation robustness in machine learning.

Future directions include extending to two-qubit measurements and non-Clifford gates (requiring tensor network methods), implementing OTOC diagnostics to directly measure scrambling, studying the transition in open systems with non-Markovian noise, and making the analogy with ML threshold phenomena more precise via representation rank diagnostics applied to large language models.

The author thanks Prof. Yi-Zhuang You for lecture material motivating this investigation, and acknowledges Ref. [9] (co-authored by Prof. You) as a foundational analytical reference for the replica field theory of the MIPT. Simulations used Claude Code (Anthropic) for AI-assisted implementation of the stabilizer tableau and circuit infrastructure.

Stabilizer Tableau Gate Update Rules

Following Aaronson and Gottesman [7] (Table 1), the gate update rules are applied simultaneously to all rows $h \in \{0, \dots, 2n - 1\}$ of the tableau. Let $x_{h,j}$ and $z_{h,j}$ denote the X - and Z -bits for qubit j in row h , and r_h the phase bit.

Hadamard on qubit a . The X - and Z -bits are swapped; the phase flips whenever both were 1:

$$\begin{aligned} r_h \oplus &= x_{h,a}^{\text{old}} \cdot z_{h,a}^{\text{old}}, \\ x_{h,a} &\leftarrow z_{h,a}^{\text{old}}, \quad z_{h,a} \leftarrow x_{h,a}^{\text{old}}. \end{aligned} \quad (5)$$

The copy of $x_{h,a}^{\text{old}}$ must be saved before any modification.

Phase (S) gate on qubit a . The phase flips when $x_{h,a} = 1$ and the old $z_{h,a} = 1$ (i.e., the generator has Y on qubit a); then $z_{h,a}$ is toggled:

$$\begin{aligned} r_h \oplus &= x_{h,a} \cdot z_{h,a}^{\text{old}}, \\ z_{h,a} \oplus &= x_{h,a}. \end{aligned} \quad (6)$$

One can verify: $S|Y\rangle \rightarrow -|X\rangle$, i.e., $Y \mapsto -X$, consistent with $SYS^\dagger = -X$.

CNOT with control a , target b .

$$\begin{aligned} r_h \oplus &= x_{h,a} \cdot z_{h,b} \cdot (x_{h,b} \oplus z_{h,a} \oplus 1), \\ x_{h,b} \oplus &= x_{h,a}, \\ z_{h,a} \oplus &= z_{h,b}. \end{aligned} \quad (7)$$

All three updates use copies of the pre-update columns; the order of the last two may be exchanged freely.

Rowsum phase function. The measurement algorithm requires computing the product of two Pauli operators. The phase contribution is [7]:

$$g(x_1, z_1, x_2, z_2) = \begin{cases} z_2 - x_2 & x_1 = 1, z_1 = 1, \\ z_2(2x_2 - 1) & x_1 = 1, z_1 = 0, \\ x_2(1 - 2z_2) & x_1 = 0, z_1 = 1, \\ 0 & x_1 = 0, z_1 = 0, \end{cases} \quad (8)$$

and the new phase bit of row h after adding row i is:

$$r_h \leftarrow \left\lfloor \frac{1}{2} \left(2r_h + 2r_i + \sum_j g_{ij} \right) \bmod 4 \right\rfloor, \quad (9)$$

where $g_{ij} \equiv g(x_{i,j}, z_{i,j}, x_{h,j}, z_{h,j})$ and the sum runs over $j = 0, \dots, n-1$. The sum runs over all n qubits and must be computed in ordinary (mod-4) integer arithmetic—not mod-2.

GF(2) Rank Algorithm

Entanglement entropy requires computing $\text{rank}_{\text{GF}(2)}(M_B)$, the rank of the binary matrix M_B over the field with two elements. **numpy.linalg.matrix_rank operates over $\mathbb{R}$ and cannot be used here:** floating-point arithmetic does not respect GF(2) arithmetic, and it returns incorrect ranks for binary matrices with linearly dependent rows that sum to zero modulo 2.

The correct algorithm is Gaussian elimination with XOR operations:

1. Copy M_B to a working matrix M of dtype `int8`.

2. Initialize `rank = 0` and `pivot_row = 0`.

3. For each column $c = 0, 1, \dots, 2|B| - 1$:

- (a) Find the first row $r \geq \text{pivot_row}$ with $M[r, c] = 1$. If none, skip to the next column.
- (b) Swap rows r and `pivot_row`.
- (c) For every other row $r' \neq \text{pivot_row}$ with $M[r', c] = 1$: set $M[r'] \oplus= M[\text{pivot_row}]$ (XOR, element-wise).
- (d) Increment `rank` and `pivot_row`.

4. Return `rank`.

XOR ($\oplus$) implements addition in GF(2) without modular reduction overhead. The algorithm runs in $O(n \cdot |B|^2)$ time, which is negligible compared to the $O(n^2)$ cost of a single measurement operation.

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